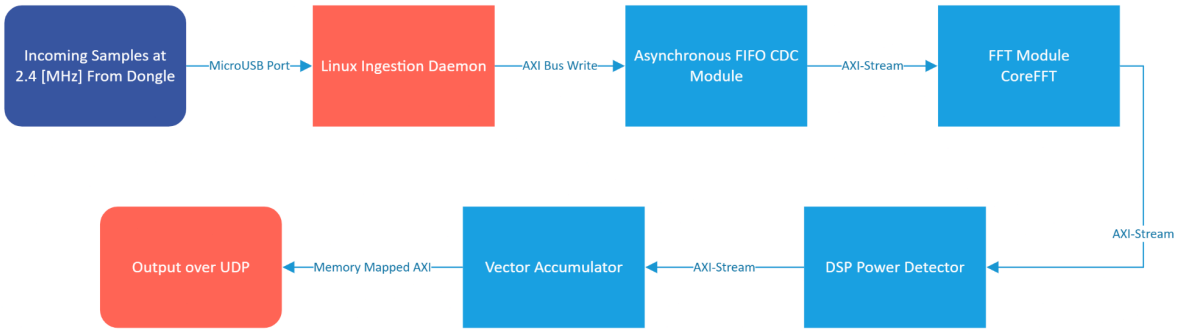
Overview

Our project is to design and implement a high-speed data interface for our university club's radio telescope to observe the 21cm hydrogen line from space.

The setup uses a PolarFire SoC Icicle Kit to receive 2.4 Msamples/sec of digitized baseband I/Q data directly from an SDR module. Because the front-end digitizer runs on its own sample clock, we feed the raw stream through a dual-clock asynchronous FIFO to safely handle the clock domain crossing into the FPGA over AXI-Stream. From there, the data feeds into a pipelined CoreFFT engine mapped into the Icicle's dedicated math blocks and LSRAM.

The FFT output is converted into real power ($I^2 + Q^2$) and compiled together in a hardware vector accumulator on fabric to pull the hydrogen line out of the thermal noise floor. Because the telescope integrates over time, the output spectrum updates about once a second so the FPGA only needs to surface a couple thousand bins at a time. The onboard 64-bit RISC-V subsystem running embedded Linux reads these integrated power bins across a memory-mapped fabric interface, handles Doppler shift conversions to map velocity profiles, and streams the spectrum out over UDP via Gigabit Ethernet to a laptop or onto the internet for real-time visualization and steering control.

System Block Diagram



Innovation & Impact

Our project tracks signals from the 1420 MHz hydrogen line, a radio frequency released by interstellar hydrogen gas clouds forming the spiral arms of the Milky Way. Using the PolarFire SoC Icicle Kit allows us to automate data acquisition and spectral processing directly in hardware. Our solution is novel because the Icicle kit draws much less power while guaranteeing zero dropped samples compared to an FFT-capable laptop, which is the standard tool of choice for accessible radio astronomy. By offloading streaming FFTs and long-term vector accumulation into the Icicle’s dedicated math blocks and LSRAM, the onboard 64-bit RISC-V cores are free to run Linux and manage our club-designed antenna steering subsystem.

Together, this creates a fully self-contained, autonomous radio telescope that continuously collects data over long periods without supervision. This extended measurement window provides ample integration for precise signal power and Doppler calculations, yielding better hydrogen density and radial velocity profiles.

The antenna assembly, steering, and Doppler software teams have drawn in over 10 students from across our city to collaborate on building the wider project, igniting a spark of interest in radio astronomy for many who heard about it for the first time. While our primary goal is to map the Milky Way, the Icicle’s high uptime and minimal data loss enable our future plan to observe faint extragalactic hydrogen signatures from the Andromeda galaxy.